Further Kinematics

Overview

Use vectors to represent motion. Can we use the *suvat* equations with constant acceleration? What changes when acceleration varies with time t?

1. Vector equations for motion

A particle has velocity

$\mathbf{v} = (1 - 2t)\mathbf{i} + (3t - 3)\mathbf{j} \quad \text{m s}^{-1}$.

Find (a) its speed at $t = 0$; (b) its bearing at $t = 2$; (c) when it moves (i) parallel to $\mathbf{j}$, (ii) parallel to $-\mathbf{i} - 3\mathbf{j}$. (3, 2, 6 marks)

2. Variable acceleration with vectors

A particle P, mass $0.8\,\text{kg}$, is acted on by a single force $\mathbf{F}$. Its position relative to fixed origin O is

$\mathbf{r} = 2t^3\mathbf{i} + 50t^{-1/2}\mathbf{j} \quad \text{m}, \quad t > 0$.

Find (a) its speed at $t = 4$; (b) acceleration as a vector at $t = 2$; (c) $\mathbf{F}$ at $t = 2$.

3. Integration for velocity/displacement

A particle moving in a plane has velocity

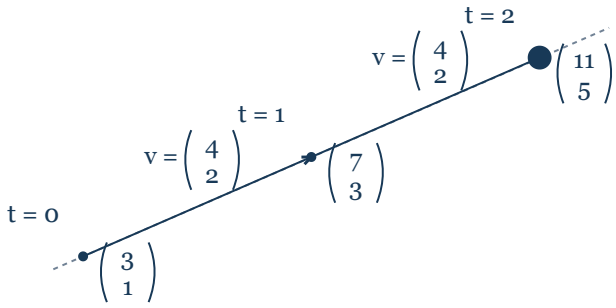
$\mathbf{v} = 3t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} \quad \text{m s}^{-1}, \quad t \geq 0$.

At $t = 0$, its position relative to O is $(2\mathbf{i} - 3\mathbf{j})\,\text{m}$. Find its position vector at time t.

Vector Motion 3

Initially, a ball has position vector $\begin{pmatrix}3\\1\end{pmatrix}$ m. Each second it moves by $\begin{pmatrix}4\\2\end{pmatrix}$ m: its constant velocity is $\begin{pmatrix}4\\2\end{pmatrix}\,\mathrm{m\,s^{-1}}$.

Where is it after 1 second? After 2 seconds?



1 second

2 seconds

In general, where is the ball after t seconds?

General

Position vector

$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$,

where $\mathbf{r}_0$ is initial position and $\mathbf{v}$ is constant velocity.

Example 4

A particle starts at $(3\mathbf{i} + 7\mathbf{j})$ m, moving with constant velocity $(2\mathbf{i} - \mathbf{j})\,\mathrm{m\,s^{-1}}$.

(a) Find its position vector 4 seconds later.

(b) Find when it is due east of the origin.

a

b

Column vectors may be easier to manipulate than i and j notation; either may be used.

SUVAT with Vectors 5

Some *suvat* equations work with vectors. Here $\mathbf{r}$ denotes displacement from the initial position:

$\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2, \qquad \mathbf{v} = \mathbf{u} + \mathbf{a}t.$

Which quantities are vectors, and which are scalars?

r

u

v

a

t: scalar. The scalar formula $v^2 = u^2 + 2as$ cannot be used by simply squaring vector symbols.

A particle has initial velocity $(-3\mathbf{i} + \mathbf{j})\,\mathrm{m\,s^{-1}}$ and constant acceleration $(2\mathbf{i} + 3\mathbf{j})\,\mathrm{m\,s^{-2}}$. Find (a) its speed and (b) its bearing at $t = 3\,\mathrm{s}$.

Speed

Bearing

Further Example 6

An ice skater is at fixed point O at $t = 0$, with velocity $(2.4\mathbf{i} - 0.6\mathbf{j})\,\mathrm{m\,s^{-1}}$. At $t = 20\,\mathrm{s}$, velocity is $(-5.6\mathbf{i} + 3.4\mathbf{j})\,\mathrm{m\,s^{-1}}$. Model the skater as a particle with constant acceleration. Find (a) acceleration; (b) position $\mathbf{s}$ relative to O at time t; (c) when directly north-east of O. A second skater has position $\mathbf{r} = (1.1t - 6)\mathbf{j}\,\mathrm{m}$. (d) Show that they meet.

a

b

c

d

Test Your Understanding 7

Edexcel M1 (Old), May 2013(R), Q6

Unit vectors $\mathbf{i}, \mathbf{j}$ point east and north. Ship S has constant velocity $(3\mathbf{i} + 3\mathbf{j})\,\text{km h}^{-1}$; at $t = 0$, position is $(-4\mathbf{i} + 2\mathbf{j})$ km. (a) Find its position at time t hours. Ship T has velocity $(-2\mathbf{i} + n\mathbf{j})\,\text{km h}^{-1}$, initial position $(6\mathbf{i} + \mathbf{j})$ km. They meet at P. (b) Find n. (c) Find OP. (2,5,4 marks)

a

b

c

Exercise 8A — Y2 Mechanics Pages 162–164 8

10. A model boat has constant acceleration $(0.2\mathbf{i} + 0.6\mathbf{j})\,\text{m s}^{-2}$. After 20 seconds its velocity is $(4\mathbf{i} + 3\mathbf{j})\,\text{m s}^{-1}$. Find displacement from its starting position after 20 seconds.

Working

11. Particle A starts at $(12\mathbf{i} + 12\mathbf{j})$ m, with velocity $(-\mathbf{i} + \mathbf{j})\,\mathrm{m\,s^{-1}}$ and constant acceleration $(2\mathbf{i} - 4\mathbf{j})\,\mathrm{m\,s^{-2}}$. Particle B has initial velocity $\mathbf{i}\,\mathrm{m\,s^{-1}}$ and constant acceleration $2\mathbf{j}\,\mathrm{m\,s^{-2}}$. They collide after 3 seconds. Find (a) both speeds then; (b) the collision point's position vector; (c) B's initial position vector.

a

b

c

12. A ship is at O at 12:00, with velocity $(-4\mathbf{i} + 8\mathbf{j})\,\mathrm{km\,h^{-1}}$. At 14:00 its velocity is $(-2\mathbf{i} - 6\mathbf{j})\,\mathrm{km\,h^{-1}}$. At t hours after 12:00, $t \geq 0$, its displacement is $\mathbf{s}$. Model constant acceleration. Find (a) acceleration; (b) $\mathbf{s}(t)$; (c) when directly south-west of O. A second ship has position $\mathbf{r} = (40 - 25t)\mathbf{j}$ km. (d) Find where they meet.

a

c

b

d

CHALLENGE — Y2 Mechanics Page 164 **11**

At an air show, an aeroplane passes over a control tower with velocity $(20\mathbf{i} - 100\mathbf{j})\,\mathrm{m\,s^{-1}}$, flying horizontally with constant acceleration $6\mathbf{j}\,\mathrm{m\,s^{-2}}$. A second plane passes the tower $t > 0$ seconds later, with velocity $(70\mathbf{i} + 40\mathbf{j})\,\mathrm{m\,s^{-1}}$, in a higher horizontal plane with acceleration $-8\mathbf{j}\,\mathrm{m\,s^{-2}}$. They subsequently pass directly over one another. Find t.

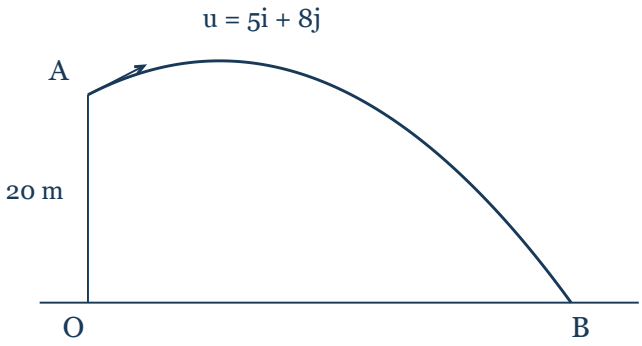
Working

Working

Vector Methods for Projectiles **12**

A velocity vector gives both direction and magnitude, so can represent projection and subsequent motion.

A racket strikes a ball at A, position $20\mathbf{j}$ m relative to O. Initial velocity is $(5\mathbf{i} + 8\mathbf{j})\,\mathrm{m\,s^{-1}}$, with $\mathbf{i}$ horizontal and $\mathbf{j}$ upward. It moves freely under gravity until hitting horizontal ground at B. Find (a) speed after 1.5 seconds; (b) position $\mathbf{r}(t)$ relative to O; (c) OB.



a

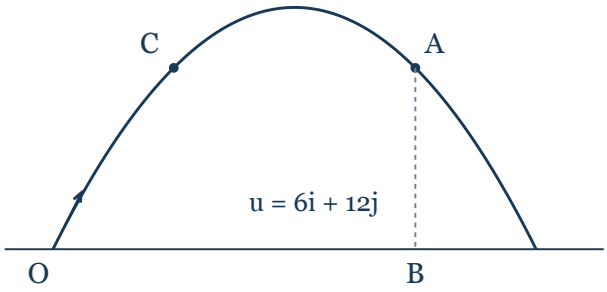
b

c

Test Your Understanding 13

Edexcel M2 (Old), January 2012, Q7

i, **j** are horizontal and vertical upward unit vectors. A ball is projected from O on horizontal ground with velocity $(6\mathbf{i} + 12\mathbf{j})\,\mathrm{m\,s^{-1}}$. It passes A at time t; B is vertically below A on the ground, with $OB = 2AB$. Find (a) t; (b) speed V at A. At another point C, speed is also V. (c) Find time from O to C. (7,5,3 marks)



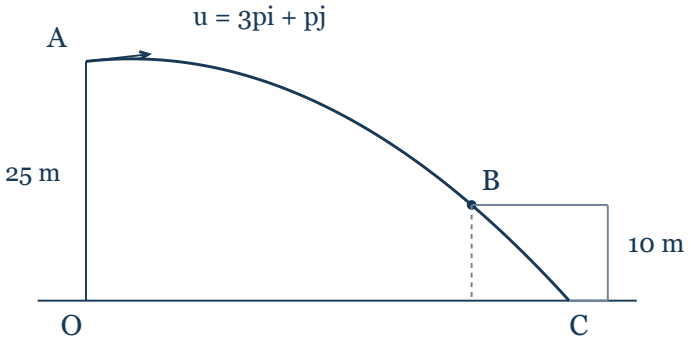
a

b

c

Exercise 8B — Y2 Mechanics Pages 166–167 14

3. A ball is projected from A atop a cliff, position $25\mathbf{j}$ m relative to base O at sea level. Initial velocity is $(3p\mathbf{i} + p\mathbf{j})\,\mathrm{m\,s^{-1}}$, with constant p. After 2 seconds it passes B, position $(q\mathbf{i} + 10\mathbf{j})$ m, then hits the sea at C. Model the ball as a particle moving freely under gravity and the sea as horizontal. Find velocity at B.



Working

Working

4. A particle P is projected from O on horizontal ground with velocity $(3u\mathbf{i} + 4u\mathbf{j})\,\mathrm{m\,s^{-1}}$. It lands 750 m from O. (a) Show $u = 17.5$. (b) Find greatest height. (c) Find the angle its direction of motion makes with $\mathbf{i}$ when $t = 5\,\mathrm{s}$.

a

b

c

5. A particle is projected from O atop a cliff with velocity $(8\mathbf{i} + 10\mathbf{j})\,\mathrm{m\,s^{-1}}$. Moving freely under gravity, it strikes the sea at S after 6 seconds. Find (a) horizontal distance OS; (b) vertical distance OS. At time T, velocity is $(8\mathbf{i} - 14.5\mathbf{j})\,\mathrm{m\,s^{-1}}$. (c) Find T and position relative to O.

a

b

c

Exercise 8B — Y2 Mechanics Pages 166–167 17

6. Use $g = 10\,\text{m s}^{-2}$. A body B is projected from O on horizontal ground with velocity $(a\mathbf{i} + b\mathbf{j})\,\text{m s}^{-1}$, where $a, b > 0$. It moves freely under gravity until landing at P, immediately coming to rest. Its position on the path is $(x\mathbf{i} + y\mathbf{j})\,\text{m}$.

(a) Show $y = bx/a - 5x^2/a^2$.

(b) Given $a = 8$, range $OP = X$ and maximum height Y, find (i) X, (ii) Y, in terms of b.

a

b.i

b.ii

Variable Acceleration in One Dimension 18

$s \xrightarrow{d/dt} v \xrightarrow{d/dt} a$

$s \xleftarrow{\int v\,dt} v \xleftarrow{\int a\,dt} a$

In Mechanics Year 1, velocity is rate of change of displacement and acceleration is rate of change of velocity:

$$v = \frac{ds}{dt}, \qquad a = \frac{dv}{dt}.$$

Reverse by integration: $v = \int a\,dt$, $s = \int v\,dt$, with constants found from initial conditions. Stay in one dimension, now using more complex Pure Year 2 functions.

A particle moving in a straight line has

$a = \cos(2\pi t)\,\text{m s}^{-2}, \quad t \geq 0.$

At $t = 0$, $v = 1/(2\pi)\,\text{m s}^{-1}$. Find (a) $v(t)$; (b) maximum speed; (c) distance travelled in the first 3 seconds.

a

b

c

Test Your Understanding 19

A particle of mass 6 kg moves on the positive x-axis. Its displacement from the origin is

$$s = 2t^{3/2} + \frac{e^{-2t}}{3}\ \text{m}, \quad t \geq 0.$$

(a) Find velocity when $t = 1.5$. A single force of magnitude F acts along positive x. (b) Find F when $t = 2$.

a

b

Chain-rule recap:

$$\frac{d}{dx}(e^{kx}) = ke^{kx}.$$

Exercise 8C — Y2 Mechanics Pages 168–170 20

7. A particle moves in a straight line with acceleration $3\sqrt{t}\,\text{m s}^{-2}$, $t > 0$, measured from when it begins motion. After 1 second, velocity is 2 m/s. Find the time taken to travel 16 m.

Working

Working

8. In a 200 m straight sprint, a runner's displacement from the start is modelled by $s = k\sqrt{t}$, $0 \le t \le T$, in metres and seconds. She finishes in 25 seconds. Find (a) k, T; (b) her speed at the finish; (c) a criticism for small t.

a

b

c

9. A particle moves in a straight line with velocity $v = 2 + 8\sin(kt)\,\text{m s}^{-1}$, $t \ge 0$. Initial acceleration is $4\,\text{m s}^{-2}$. (a) Find k. Using it, (b) find all $t \in [0, 4\pi]$ when $a = 0$; (c) show $4a^2 = 64 - (v-2)^2$; (d) find maximum velocity and maximum acceleration.

a

b

c

d

Exercise 8C — Y2 Mechanics Pages 168–170

10. A particle moves along the x-axis, with velocity in the positive direction

$$v = \begin{cases} 10t - 2t^{3/2}, & 0 \le t \le 4, \\ 24 - \left(\frac{t-4}{2}\right)^4, & t > 4. \end{cases}$$

At $t = 0$, it is at the origin. Find (a) greatest speed on $[0, 4]$; (b) distance from O at $t = 4$; (c) when at rest for $t > 4$; (d) total distance in the first 10 seconds.

a

b

c

d

Differentiating Vectors

If $\mathbf{v} = \begin{pmatrix} t^2 \\ \sin t \end{pmatrix}$, what is the acceleration?

Derivative

If $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j},$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2} = \ddot{\mathbf{r}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}.$$

Dot notation means differentiation with respect to time, e.g. $\dot{x} = dx/dt$; it is common in Physics.

A particle P, mass 0.8 kg, is acted on by a single force $\mathbf{F}$. Its position is

$\mathbf{r} = 2t^3\mathbf{i} + 50t^{-1/2}\mathbf{j}$ m, $\quad t > 0$.

Find (a) speed at $t = 4$; (b) acceleration vector at $t = 2$; (c) force at $t = 2$.

a

b

c

4. Position relative to fixed O is

$\mathbf{r} = 4t^2\mathbf{i} + (24t - 3t^2)\mathbf{j}\,\text{m}, \quad t \geq 0.$

(a) Find speed at $t = 2$. (b) Show acceleration is constant and find its magnitude.

a

b

5. Particle P, mass 0.5 kg, starts at O, projected at $t = 0$. Its position is

$\mathbf{r} = (t^3 - 12t)\mathbf{i} + (4t^2 - 6t)\mathbf{j}\,\text{m}, \quad t \geq 0.$

Find (a) projection speed; (b) when moving parallel to $\mathbf{j}$; (c) position then. A single variable force $\mathbf{F}$ causes the motion. (d) Find $|\mathbf{F}|$ at $t = 5$.

a

b

c

d

6. A particle moving in a plane has position

$\mathbf{r} = (3t^2 - 6t + 4)\mathbf{i} + (t^3 + kt^2)\mathbf{j}$ m.

At $t = 3$, its speed is $12\sqrt{5}\,\text{m s}^{-1}$. (a) Find both possible k. (b) For each, find acceleration magnitude at $t = 1.5$.

a

b

b

7. A particle's position is

$\mathbf{r} = 6t^2\mathbf{i} + t^{5/2}\mathbf{j}$ m, $\quad t \geq 0$.

At $t = 4$, find (a) speed; (b) acceleration as a vector.

a

b

8. A particle moves in a horizontal plane with

$\mathbf{r} = (18t - 4t^3)\mathbf{i} + ct^2\mathbf{j}\ \text{m}, \quad t \geq 0,\ c > 0.$

At $t = 1.5$, speed is 15 m/s. Find (a) c; (b) acceleration at $t = 1.5$.

8a

8b

9. A particle has position

$\mathbf{r} = (2t^2 - 3t)\mathbf{i} + (5t + t^2)\mathbf{j}\ \text{m}, \quad t \geq 0.$

Show acceleration is constant and find its magnitude.

9

10. A particle moves in a horizontal plane with

$\mathbf{r} = (20t - 2t^3)\mathbf{i} + kt^2\mathbf{j}\ \text{m}, \quad t \geq 0,\ k > 0.$

At $t = 2$, speed is 16 m/s. Find (a) k; (b) acceleration when moving parallel to $\mathbf{j}$.

a

b

Integrating Vectors 31

Integrate vector components independently. A particle moving in a plane has velocity

$\mathbf{v} = 3t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} \quad \text{m s}^{-1}, \quad t \geq 0.$

At $t = 0$, position relative to O is $(2\mathbf{i} - 3\mathbf{j})$ m. Find its position vector at time t.

Integrate

Position

Further Example 32

A particle has acceleration $\mathbf{a} = 4\mathbf{i} - 2t\mathbf{j}\,\text{m s}^{-2}$. At $t = 3$, velocity is $6\mathbf{i}\,\text{m s}^{-1}$ and position is $(20\mathbf{i} + 3\mathbf{j})$ m. Find (a) direction of motion at $t = 2$; (b) distance from O at $t = 0$.

a

b

Test Your Understanding 33

Edexcel M2 (Old), January 2013, Q4

Particle P has velocity $(4t - 5)\mathbf{i} + 3\mathbf{j}\,\text{m s}^{-1}$, and initial position $(2\mathbf{i} + 5\mathbf{j})\,\text{m}$. Find (a) when moving parallel to $\mathbf{j}$; (b) its position at time t. Particle Q has constant velocity $(-2\mathbf{i} + c\mathbf{j})\,\text{m s}^{-1}$, initial position $(11\mathbf{i} + 2\mathbf{j})\,\text{m}$. They collide at $(d\mathbf{i} + 14\mathbf{j})\,\text{m}$. Find (c)(i) c, (ii) d.

a

b

c

Alternative

Exercise 8E — Integrating Vectors 34

1. A particle has acceleration
   $\mathbf{a} = 6t^2\mathbf{i} + (8 - 4t^3)\mathbf{j}\,\text{m s}^{-2}$.
   It is initially at rest at O. Find (a) velocity at $t = 2$; (b) position at $t = 4$.

a

b

2. A particle has velocity

$\mathbf{v} = (3t^2 + 2)\mathbf{i} + (6t - 4)\mathbf{j}\,\mathrm{m\,s^{-1}}, \quad t \geq 0.$

At $t = 2$, its position is $9\mathbf{j}$ m. Find (a) its initial distance from O; (b) its acceleration when moving parallel to $\mathbf{i}$.

a

b

3. A particle has acceleration $\mathbf{a} = (2t - 4)\mathbf{i} + 6\sin t\,\mathbf{j}\,\mathrm{m\,s^{-2}}$. It is at rest when $t = \pi/2$. Find (a) velocity at time t; (b) exact speed at $t = 3\pi/2$.

a

b

4. A particle has acceleration $\mathbf{a} = (5t - 3)\mathbf{i} + (8 - t)\mathbf{j}\,\text{m s}^{-2}$ and initial velocity $(2\mathbf{i} - 5\mathbf{j})\,\text{m s}^{-1}$. Find (a) velocity at time t; (b) when moving parallel to $\mathbf{i} - \mathbf{j}$; (c) its speed then.

a

b

c

5. Particle P has acceleration $2\mathbf{i} - 2t\mathbf{j}\,\text{m s}^{-2}$, initial velocity $2\mathbf{j}\,\text{m s}^{-1}$ and initial position $6\mathbf{i}\,\text{m}$ relative to fixed O. (a) Find its position at time t. Particle Q has velocity $(3t^2 - 4)\mathbf{i} - 2t\mathbf{j}\,\text{m s}^{-1}$. They collide at $t = 3$. (b) Find Q's initial position.

a

b

Exercise 8E — Integrating Vectors 39

6. A particle starts from rest at position $(4\mathbf{i} - 6\mathbf{j})$ m. Its acceleration is $(4t - 3)\mathbf{i} - 6t^2\mathbf{j}\,\mathrm{m\,s^{-2}}$. Find (a) velocity at $t = 1/2$; (b) position at $t = 6$.

a

b

Exercise 8E — Integrating Vectors 40

8. Particle P has velocity $(4t - 3)\mathbf{i} + 4\mathbf{j}\,\mathrm{m\,s^{-1}}$ and initial position $(\mathbf{i} + 2\mathbf{j})$ m. (a) Find its position at time t. Particle Q has velocity $(5\mathbf{i} + k\mathbf{j})\,\mathrm{m\,s^{-1}}$ and initial position $(11\mathbf{i} + 5\mathbf{j})$ m. They collide. Find (b)(i) k, (ii) the collision position.

a

b.i

b.ii